Variational Autoencoder for Chest X-Ray Images

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Abstract

In this work, we implement a variational autoencoder (VAE) [3] for chest X-Ray images. We report per-pixel MSE between inputs and reconstructions, FID [1] between real and reconstructed images, and Inception Score [7]. Additionally, we report FID and Inception Score for randomly sampled images. Code can be accessed on GitHub: `https://github.com/jonathansickert/xray-vae`.

1 Data

We use the public Chest X-Ray Images (Pneumonia) [4] from Kaggle, consisting of 5,863 gray-scale images from two classes: NORMAL and PNEUNIBUA. Before the images are passed to the VAE, we resize every image to 224×224 resolution and rescale to $[-1, 1]$. The dataset already has predefined train, validation and test splits.

2 Method

We implement the β-VAE [2] variant, composed of an encoder $\mathcal{E}$ and decoder $\mathcal{D}$. The encoder is composed of 5 2D convolutional blocks with channels $1 \to 32 \to 64 \to 128 \to 256 \to 256$, interleaved with batch norm and leaky ReLU (`negative_slope` $= 0.2$). Each convolution uses a 4×4 kernel with stride 2 and padding 1, so that a $1 \times 224 \times 224$ input is mapped to a $256 \times 7 \times 7$ feature map. This feature map is flattened and passed through two separate linear layers that predict the mean $\mu \in \mathbb{R}^d$ and log-variance $\log \sigma^2 \in \mathbb{R}^d$. A latent code is sampled via the reparameterization trick, $z = \mu + \sigma \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$.

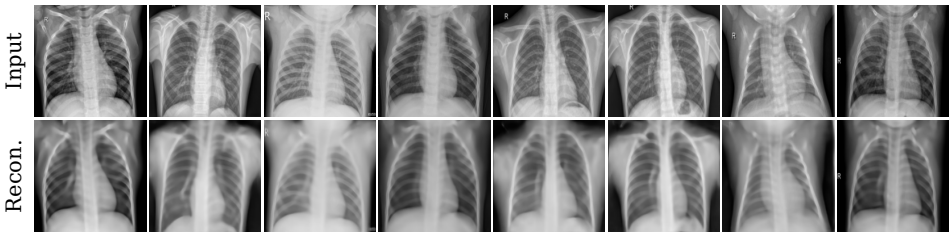
The decoder first projects z with a linear layer to $256 \cdot 7 \cdot 7$ dimensions, which are reshaped into a $256 \times 7 \times 7$ feature map and refined by two residual blocks. Each residual block consists of two 3×3 convolutions with batch norm, separated by a ReLU, followed by an identity skip connection and a final ReLU. The feature map is then upsampled to full resolution by 5 upsampling blocks with channels $256 \to 256 \to 192 \to 128 \to 64 \to 32$. Each upsampling block applies nearest-neighbor upsampling by a factor of 2, a 3×3 convolution with batch norm and ReLU, and two residual blocks (one in the final block), increasing the spatial resolution from 7×7 to 224×224. Finally, a 3×3 convolution maps the features to a single channel, followed by a final $\tanh$ activation.

We train the VAE by minimizing a β-weighted negative ELBO [2],

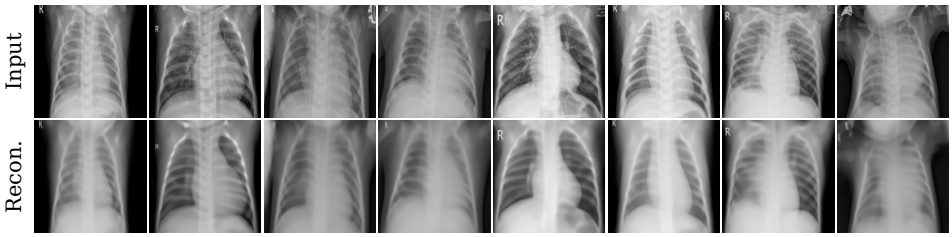
$$\mathcal{L} = \|x - \mathcal{D}(z)\|_2^2 + \beta \, D_{\mathrm{KL}}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, I)\big) \,, \tag{1}$$

where the reconstruction error is summed over pixels and averaged over the batch. We linearly increase β from 0 to 0.5 over the first 10 epochs. We use Adam with a learning rate of 5×10^{-4}, decayed to 10^{-6} with cosine annealing over 100 epochs, a batch size of 32, and a latent dimension of $d = 128$. All networks are implemented using the PyTorch library [6].

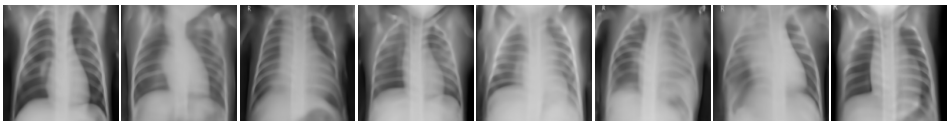
Preprint.



(a) NORMAL: input (top) and reconstruction (bottom)

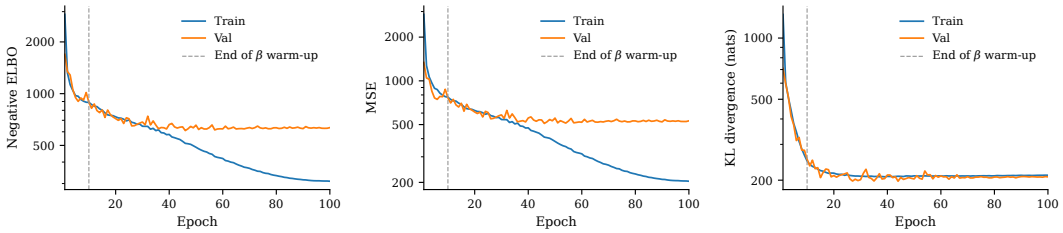


(b) PNEUMONIA: input (top) and reconstruction (bottom)



(c) Random samples decoded from $z \sim \mathcal{N}(0, I)$

Figure 1: Reconstructions of eight randomly chosen test images per class, and eight random samples.



(a) Training objective (b) Reconstruction error (c) KL divergence

Figure 2: Training curves on the training and validation sets.

3 Evaluation

We evaluate the model on the 624 images of the test set using the per-pixel MSE between inputs and reconstructions, the Fréchet Inception Distance (FID) [1] against the real test images, and the Inception Score (IS) [7], all computed with `torchmetrics` [5]. Additionally, we evaluate random samples obtained by decoding $z \sim \mathcal{N}(0, I)$ with $\mathcal{D}$. Qualitative results are shown in Figure 1, and training curves in Figure 2.

Reconstructions reach a per-pixel MSE of 0.0119 and an FID of 125.4. Random samples reach an FID of 128.2, indicating that decoded prior samples are of comparable quality to reconstructions. The IS of reconstructions (1.98) and samples (2.02) is close to that of the real images (2.25).

4 Conclusion

Our β-VAE produces faithful reconstructions and random samples of comparable quality. The training curves show overfitting: training error keeps decreasing while validation error plateaus, which early stopping or data augmentation could address. Choosing $\beta = 0.5$ favors reconstruction quality, and the KL term remains above zero, so no posterior collapse occurs. Perceptual or adversarial losses could further sharpen the outputs.

References

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